



Wireless Charging Of Electric Cars: Close To Reality

When chargers embedded into the roads become a commercial viability, the impact would be revolutionary and crucial for autonomous cars of the future. Instead of taking the EVs to the charger every few hours, these would be constantly charged from the road as they drive along



An electric car being charged using a bulky cable (Credit: www.pocket-lint.com/cars)

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In conventional cars, travelling a long way is not a problem with easy and quick refuelling. However, in today's electric cars, extended journeys require careful planning to ensure that the battery is fully charged at the starting point and that compatible fast-charging stations are available en route. It would be much more convenient if batteries could be recharged wirelessly while travelling on the road so that no break for recharging is necessary.

Consider a future scenario wherein, as you exit a building, a driverless ridesharing electric vehicle (EV) pulls over, takes you to your destination and then proceeds to drive passenger after passenger, without ever

stopping to recharge its battery. Instead, the renewable power generated by nearby wind or solar resources is delivered wirelessly to the roadway to recharge while the EV is moving.

Not needing to stop for recharging will make EVs truly autonomous, enabling them to remain in service for more hours, and thus fewer vehicles will be needed to meet the same passenger demand. Also, their batteries could be made much smaller, leading to reduced costs and increased adoption. Basically, wireless vehicle charging, much like placing a smartphone on a charging pad instead of plugging it in, charges the battery when the car is parked over a charger pad placed on the ground below. This means no bulky cables need to be lifted out of the boot of the vehicle.

In fact, there is no requirement of cables in the park and charge system. The drivers simply need to pull into a parking space to charge the batteries, start and end the charging using their smartphones, and pay for the services rendered using an app. Inductive charging based on electromagnetic induction is used to charge the EV batteries wirelessly, much like the other electric devices with wireless charging facility, such as mobile phones, electric toothbrushes, etc.

In inductive charging, electric energy gets transferred through an air gap from one magnetic coil to another when the two coils are aligned with each other. An alternating current from an external source is made to flow through a coil to create a varying magnetic field, which then using electromagnetic induction generates alternating current in the other